WEEK 3 · LESSON 13 OF 24 · HANDOUT

Training, Testing and Overfitting

Study time: **about 45 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=EuBBz3bI-aA — open the lesson online to watch inside the course.

🎓 **Go deeper:** Harvard CS50 AI, Lecture 4: Learning (supervised, reinforcement, unsupervised) — https://www.youtube.com/watch?v=E4M_IQG0d9g

Your goal in this lesson: Explain why data is split into training, validation and test sets, recognise overfitting and underfitting, and name ways to prevent overfitting.

Memorising is not learning

Imagine a student who memorises the answers to last year's exam word for word. On last year's paper they score 100%. On this year's paper, with new questions, they fail. They memorised — they did not learn.

Models can do the same thing. A model that is perfect on its training data may be useless on new data. To know the truth, we must test it on examples it has **never seen**.

Three sets of data

Set	Typical share	Used for
Training set	about 70%	The model learns its patterns here
Validation set	about 15%	Compare models and tune settings during development
Test set	about 15%	One final, honest check at the very end

The test set is like the sealed national exam paper: you open it **once**. If you keep checking it and changing the model, you slowly "teach to the test", and the score stops being honest.

Cross-validation is a way to use small datasets well: split the data into, say, 5 parts, train on 4 and validate on 1, rotate five times, and average the results.

Overfitting and underfitting

	Underfitting	Good fit	Overfitting
What happens	The model is too simple to capture the pattern	Captures the real pattern	Learns the noise and accidents in the training data
Training score	Poor	Good	Excellent
New-data score	Poor	Good	Poor
Exam analogy	Did not study	Understood the topic	Memorised last year's answers

The tell-tale sign of **overfitting** is a big gap: very high training score, much lower validation score.

Why overfitting happens

- The model is **too complex** for the amount of data (a very deep decision tree on 200 rows).
- **Too little data**, so random accidents look like patterns.
- **Training too long**, so the model starts memorising.
- **Noisy labels** that the model tries to copy exactly.

How to prevent it

- **Get more (good) data** — the most reliable fix.
- **Use a simpler model**, or limit its complexity (for example, limit tree depth).
- **Regularisation** — add a penalty for complexity, so the model prefers simpler patterns.
- **Early stopping** — stop training when the validation score stops improving.
- **Dropout** (in neural networks) — randomly switch off some neurons during training so the network cannot rely on any single path.

The bias–variance idea

- **High bias** = the model's assumptions are too simple → underfitting.
- **High variance** = the model changes a lot with small changes in the data → overfitting.

Good models balance the two. (This "bias" is a statistics word; it is different from unfair bias, which you study in Week 4.)

Remember this: Always judge a model on data it has not seen. A large gap between training and validation scores means overfitting; more data, simpler models, regularisation and early stopping help.

Name: _____ Date: _____

Activity

Task 1

Training accuracy 99%, validation 71%.

Task 2

Training accuracy 62%, validation 60%.

Task 3

Training accuracy 88%, validation 86%.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. Why do we keep a separate test set?
   - A. To store backups
   - B. To make training faster
   - C. Because training data is too small
   - D. To get one honest final check on data the model has never seen
2. Training accuracy 99%, validation accuracy 71%. What is the likely problem?
   - A. Data cleaning
   - B. Overfitting
   - C. A perfect model
   - D. Underfitting

3. Training 62%, validation 60%. What is the likely problem?

A. Nothing, it is perfect
B. Leakage
C. Overfitting
D. Underfitting — the model is too simple

4. Which is a way to reduce overfitting?

A. Train for much longer
B. Test on the training set
C. Get more good data or use regularisation
D. Use a more complex model on less data

5. What is early stopping?

A. Stopping the project early
B. Stopping training when the validation score stops improving
C. Deleting the test set
D. Turning off the computer

6. What is the purpose of the validation set?

A. Comparing models and tuning settings during development
B. Training the model
C. Storing labels
D. Final exam only

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